Mapping glacier retreat with satellite radar and machine learning

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Glaciers are retreating faster than at any time in the observational record, yet most of the world's glaciers are not monitored in the field. Here we show that radar images from satellites can be turned into monthly maps of glacier fronts with an accuracy close to that of field surveys. We trained a network on eleven thousand hand-drawn fronts and applied it to every glacier in the Himalaya, finding that the rate of retreat has doubled since the year two thousand and that small glaciers are the most at risk over the coming decades.

Large language models are now used to read scientific documents, yet most submission systems still ask authors to type the same metadata by hand. Prior work by Smith et al. [3] studied header parsing with conditional random fields, while Lee and Wong [5] relied on font statistics. We compare layout rules with learned models on manuscripts from many templates and report where each one fails. Large language models are now used to read scientific documents, yet most submission systems still ask authors to type the same metadata by hand. Prior work by Smith et al. [3] studied header parsing with conditional random fields, while Lee and Wong [5] relied on font statistics. We compare layout rules with learned models on manuscripts from many templates and report where each one fails.

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Fig. 1 | **Glacier fronts over time.**

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Methods

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